Po-Wei (Brian) Chen

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EDUCATION

Soochow University, Taipei, Taiwan
B.B.A. in Financial Engineering and Actuarial Mathematics — Sept. 2022 – June 2025

- Graduated one year early (completed all required credits and ranked in the top 10% of the class each semester)
- Cumulative GPA 4.0/4.0; Average 95.2/100; Class Rank 1st/53

WORK EXPERIENCE

Research Assistant (Full-time)
Institute of Statistical Science, Academia Sinica, Taipei, Taiwan — Aug. 2025 – present
PI: Prof. Hsin-Cheng Huang, Distinguished Research Fellow

- Conducted research on probabilistic generative models to improve predictive inference in limited-data settings and am currently exploring conformal prediction for spatial and spatio-temporal data.
- Managed administrative tasks and assisted with the organization of international conferences.

High School Math Tutor — Aug. 2024 – June 2025
Taiwan TACT Education (Private Tutoring Center), Minsheng Branch, Taipei, Taiwan

- Taught high school mathematics and prepared students for university entrance examinations.

HONORS & AWARDS

ISI-ISM-ISSAS Joint Conference Poster Competition Award (2026)
Title: Bayesian Inference for Diffusion Models: Integrating Synthetic and Observed Data with Catalytic Priors.

Outstanding Graduating Student (2025)
Soochow University, Taipei, Taiwan

Academic Excellence Award (2022–2025; six-time recipient; awarded to top three of 50+ students each semester)
Soochow University, Taipei, Taiwan

RESEARCH PROJECTS

Bridging Synthetic and Real Data with Catalytic Priors for Conditional Generative Models
Institute of Statistical Science, Academia Sinica, Taipei, Taiwan

- Augmented diffusion-based generative models with a Bayesian framework incorporating catalytic priors that leverage simpler statistical structures to guide learning in limited-data settings.
- **My contribution:** Conducted simulation experiments to evaluate improvements in predictive accuracy and uncertainty quantification across different data settings.

Empirical Asset Pricing via Machine Learning for Cryptocurrency
Soochow University, Taipei, Taiwan

- Proposed a portfolio construction framework for cryptocurrency markets using machine learning predictions, linking predicted returns to financial characteristics through cross-sectional rankings.
- **My contribution:** Contributed to framework development, implemented and evaluated multiple machine learning models, and conducted empirical analyses comparing the proposed framework with benchmark strategies.

MANUSCRIPTS IN PREPARATION

- "**Bridging Synthetic and Real Data with Catalytic Priors for Conditional Generative Models**" *14th IASC-ARS & SoAI Joint Conference*, Chiang Mai University, Chiang Mai, Thailand. Dec. 2026 [Accepted for Oral Presentation]
- "**Bayesian Inference for Diffusion Models: Integrating Synthetic and Observed Data with Catalytic Priors**" *ISI-ISM-ISSAS Joint Conference*, Academia Sinica, Taipei, Taiwan. Feb. 2026 [Poster]
- "**Empirical Asset Pricing via Machine Learning for Cryptocurrency**" *Annual Conference and International Symposium of the Taiwan Risk and Insurance Association*, Tamkang University, New Taipei, Taiwan. Dec. 2025 [Oral]

MANUSCRIPTS IN PREPARATION

- **Po-Wei Chen**, Yeh-Ting Chang, and Hung-Wen Cheng. *Empirical Asset Pricing via Machine Learning for Cryptocurrency*. In preparation.
- Tzu-Chi Wang, **Po-Wei Chen**, Wen-Ting Wang, Sheng-Li Tzeng, and Hsin-Cheng Huang. *Bridging Synthetic and Real Data with Catalytic Priors for Conditional Generative Models*. In preparation.

COURSEWORK & CERTIFICATION

Selected Coursework
Soochow University, Taipei, Taiwan

- **Mathematics & Analysis:** Advanced Calculus (Introduction to Analysis), Calculus I–II, Linear Algebra I–II
- **Probability & Statistics:** Probability and Statistics I–II, Statistical Inference, Advanced Statistics, Applied Statistics, Regression Analysis
- **Computational & Optimization Methods:** Numerical Analysis, Operations Research, Computer Programming I–II, C++ Programming Language, Artificial Intelligence in Finance
- **Actuarial & Financial Mathematics:** Fundamentals of Actuarial Mathematics, Mathematics of Compound Interest, Financial Calculation

Online Courses

- Python for Everybody Specialization, University of Michigan (via Coursera)
- Programming for Business Computing in Python (1), National Taiwan University (via Coursera)

Certification

- Actuarial Exam P (Probability), Society of Actuaries (SOA) — Passed

TECHNICAL SKILLS

Programming: Python, R (R Markdown, Shiny), C++, SQL
Machine Learning Packages: PyTorch, TensorFlow, scikit-learn

EXTRACURRICULAR ACTIVITIES

Statistics Cup Basketball Tournament Champions Nov. 2025
Institute of Statistical Science, Academia Sinica, Taipei, Taiwan

Vice Captain of Basketball Team 2024 – 2025
Department of Financial Engineering and Actuarial Mathematics, Soochow University, Taipei, Taiwan

- Led the team to six championships in interdepartmental competitions.